

4 Walls

What this is

4 Walls is a portfolio you walk around in.

It is a small 3D game built to look like it shipped on a PlayStation 1 disc in 1998, and all of it happens inside one square room: a twenty-year-old computer science student's bedroom in Jammu, frozen at dusk. You play as V. You can walk to the desk, switch on the monitor, and find the portfolio sitting there in a fake early-2000s desktop. You can also ignore the monitor entirely, boil a cup of chai, put on the headphones, drop into the bean bag, and listen to whatever is playing on the boombox until you get bored and close the tab.

Both of those are correct ways to play.

The usual portfolio hands you a name, a photograph, and a grid of logos, and asks you to infer a person from it. This one skips the inference. It puts you in the room and lets you open the fridge.

The name

"Within these four walls" is an idiom about privacy. It means what happens here does not leave. The room is full of things that carry exactly that weight: bills stacked on the fridge, a diary by the bed, an old phone in a wardrobe drawer, a photograph tucked between the pages of a book. The title says quietly that this is the private version, the one that does not go on a resume.

It is also just literally true. There are four walls, one door, two windows, and you never leave.

The logo

The wordmark is **W4LLS**.

1. The 4 standing in for the A is leetspeak, which was the native handwriting of the internet in the era the game is dressed as, and of the kind of person who would build this room in the first place.
2. The doubled outline sits slightly out of register, and the letters do not share a clean baseline. That misalignment is the same vertex wobble the game's shader produces on every model in the room. The logo misbehaves the way the world misbehaves.

The philosophy

There is no ending.

No missions, no objectives, no percentage in the corner, nothing to complete, no state you can reach where the game tells you that you are finished. You arrive, you poke at things, and you leave whenever you feel like leaving. Nothing is gated behind anything else, and nothing is withheld to make you work for it.

This is a deliberate choice and it costs something. Games with objectives are stickier, and a completion counter would probably keep people in the room longer. But a checklist would

turn the room into a task, and the entire point is that it is a place. People linger in places because they want to, and the moment a visitor senses they are being scored, the room stops being somewhere to sit and becomes something to clear.

So the design has to earn attention through texture instead of structure. Almost everything you can see, you can touch, and almost everything you touch does something small and specific. The reward for exploring is the exploring.

How it looks

The room is faceted, flat-shaded, low-poly, and slightly wrong on purpose.

Vertices snap to a coarse grid, so edges shimmer and jitter as the geometry moves. Textures are mapped without perspective correction, so surfaces warp and swim when viewed at an angle, the way PS1 floors always did. The whole scene renders to a low-resolution target and gets scaled up, which gives everything chunky, honest pixels. Textures are tiny, 64 or 128 pixels square. No anti-aliasing. Colour depth is limited and dithered.

The reference points are the original PlayStation and the GTA III and Vice City era: short draw distances, warm haze, texture warp, geometry that visibly struggles.

None of this is meant to look broken. There is a real difference between retro that has been designed and retro that is just low effort, and the difference lives entirely in the discipline of the texture work and the lighting. The wobble is a style, so it has to be consistent, and everything on top of it has to be clean.

The palette is warm the whole way through, because of the light.

Dusk

The room is set at golden hour and it stays there.

The sun is low outside the window, the sky runs amber near the horizon and bruises into indigo higher up, and the light comes into the room sideways and orange. Long shadows. Dust visible in the shafts. When the interior lights are on they are warm white, never cold, because a cold bulb would fight the sunset instead of sitting inside it.

Dusk is the default state, not a moment the clock happens to pass through. There is no real-time cycle running in the background, and that is on purpose: a live cycle means most visitors would never see the light the room was designed around, and it would either tick fast enough to feel like a gimmick or slow enough that nothing ever changed. Instead there are four hand-tuned states, and you move between them by sleeping.

Dusk, then night, then dawn, then midday, then back to dusk. Four lighting setups authored to look good individually, which is cheaper and better than interpolating through a hundred that were never checked.

Weather rolls once when the page loads and is heavily weighted towards clear. Rain is the uncommon one. Rain at dusk, orange light through wet glass, is the version people will screenshot.

Starting up

The disc boots.

A PS1-style startup sequence plays, because the site needs a loading screen anyway and this is the one place where the aesthetic can announce itself before anything is rendered. A start prompt. Then the room fades in at dusk.

Roughly five seconds later, the phone on the desk rings.

That call is doing three jobs at once. Answering it is a click, which is what browsers need before they will allow any audio to play, so the ringing phone unlocks the entire soundscape without a mute button ever appearing on screen. The voice on the other end is a friend asking what you are up to, and in the course of that conversation you are told how to move and how to touch things, so there is no tutorial overlay and no floating key prompts on load. And it establishes the phone as the thing that talks to you, which matters later, because occasional calls are how the room nudges you towards corners you have not looked in yet.

You are never obliged to answer. The phone stops ringing on its own if you ignore it, and the room lets you.

Seeing and moving

There are six cameras.

Four are mounted in the corners of the room, one hangs from the centre of the ceiling, and one is V's own eyes. The corner and ceiling cameras are fixed. As you move, the view cuts to whichever one frames you best, in the style of the survival horror games that shared this hardware. You can also cycle them manually.

Fixed cameras are the right answer here for two reasons. They are authentic to the era, and every shot can be composed by hand instead of fought with. A third-person camera chasing a character around a cramped bedroom is one of the genuinely hard problems in 3D, full of wall clipping and occlusion and lurching corrections, and fixed angles delete the problem instead of solving it.

But a corner camera is too far away to read a sticky note. So the first-person view is the inspection mode: get close to something and lean in, and the game drops into V's eyes so you can actually read the handwriting on a bill, the spine of a book, the words on a poster. Back out and the fixed cameras take over again. Wide shots for atmosphere and movement, V's eyes for detail.

Movement is click-to-move. Tap the floor and V walks there, tap an object and he interacts with it. Keyboard controls exist on desktop for anyone who wants them, and are never required. This means there is no separate mobile version and no virtual joystick pinned over the corner of the screen, because the input model is identical everywhere. The art style helps here too: 64-pixel textures and a low-resolution render target are not a compromise made for phones, they are the look, so phones run it comfortably.

The desk

The desk runs along the top wall under the second window, with the chair in front of it.

There is one monitor and one laptop. They are different machines with different jobs, and the split is the spine of the whole project.

The monitor is the public one. It is off when you arrive, dark glass holding a faint orange reflection of the window behind you. Switch it on and it boots into a chunky early-2000s desktop with a GitHub icon on it. Open that and the profile is there, nikhilvirdo, and from the profile you get the projects as logos rather than as a list of names, plus the tech stack, plus a short strip of things worth mentioning. Click any project logo and it takes you to the repository or the live site. That is the whole portfolio, and a visitor in a hurry can be in and out of it inside a minute.

The laptop is the private one. It holds the log of this room being built, written while it was being built, which is a better "about" than a paragraph in the third person could ever be. It holds a fake terminal, and the terminal answers a specific command with something you will only find if you go looking. And it is where you read the contents of anything you plug into it.

Which brings us to the drives.

The USB sticks scattered on the desk are not clutter and they are not project files. They are personal archives. One has the first program V ever wrote, badly. One has an old playlist. One has a folder of saved memes. One has half-finished ideas that never went anywhere. Plug a stick into the laptop and you can read what is on it.

The dead external hard drive is the good one. Plug it in and it clicks, struggles, spins up, half-mounts, and recovers exactly one corrupted file before giving up. What it recovers is something that was lost. A drive that fails in front of you is more affecting than a drive that works, and every person who has owned a hard drive for five years knows that sound.

The mechanical keyboard is white-backlit and clean, with no colour cycling. Individual keys physically depress when you click them and each one clacks. The lighting breathes very slowly. Type a particular word on it and something happens.

The mouse sits beside it, nudgeable, its cable dragging when it moves, with a single steady glow underneath. It was going to have colour cycling and does not, because at this resolution a mouse occupies about twenty pixels and an RGB gradient across it resolves to two pixels quietly changing hue. There is nothing to see, so there is nothing there.

The headphones hang off the corner of the monitor, and they are the single best interaction on the desk. Put them on and the entire audio mix changes: the room's ambience muffles and drops back, the music from the boombox comes close and intimate, footsteps go dull, the outside world thins out. Take them off and the room rushes back in around you. One object, one action, and the whole sensory character of the space flips.

The tennis ball, the soft gully cricket kind, sits on the desk where it has no business being. Pick it up and throw it. It bounces off the walls and the almirah and comes to rest wherever

it lands, and it stays there. It has no purpose. Every good room needs one thing you can throw.

The sticky notes are pinned around the desk edge and stuck to the monitor bezel. Each one is readable up close and each one has something real on it, an algorithm half-remembered, an idea for a project, a thought at two in the morning. You can peel them off and stick them somewhere else. You can also write your own: select a blank one and type, and it stays where you put it. One or two of the existing notes point at something else in the room for anyone paying attention.

The energy drink can is empty and crushable and throwable. **The tangled charging cables** shift as a mass when you tug at them, and one of them will charge the phone. There are ring stains on the desk from a mug, and the room adds new ones: set a full cup down on the wood and it leaves another. Those marks accumulate from your own play, which is a very small thing that almost nobody consciously notices and everybody registers.

The phone lies face up next to the keyboard. It shows the time. It lights up on its own with notifications now and then, which is the room's quietest idle animation and one of its most convincing. You can pick it up and call a handful of contacts who have their own things to say. It rings with alarms. It rings with incoming calls, including the one that starts the game and the ones that nudge you later. It charges if you plug it into the wall.

The bed corner

The bed is against the left wall, under the second window, with a plant beside the headboard.

You can sleep, which is how time moves: fade to black, and the room comes back in a different one of its four lighting states. You can also just lie down and look at the ceiling fan, which does nothing at all and is the correct use of a bed in a game with no objectives.

The plant by the window can be watered. It visibly straightens up when you do. It also remembers, so if you come back to the room days later it has drooped again and needs watering again, which is a very small relationship to have with a website.

The diary lives on the bedside, and it is doing three jobs. Its cover is plastered in stickers, Linux and GitHub and Python and Node and one W4LLS. Inside, the front half is a diary, handwritten, personal. The back half is where it stopped being a diary and became a DSA notebook, and it is open to a page of pseudocode for topological sort using Kahn's algorithm. That page is not static. Look at it in first person and the algorithm runs on the small graph sketched in the margin: in-degrees ticking down, the queue draining, the ordering resolving one node at a time.

That is the difference between claiming you know data structures and leaving a working one lying by the bed for someone to find.

Somewhere further back in the diary are the credits: what the room was built with, what was hard, what was borrowed. Anyone who reads that far has earned it.

The wardrobe

The almirah stands between the bed and the fridge on the left wall. It opens. Clothes on hangers, folded things on shelves, drawers that pull out, and inside the drawers the kind of objects that get put in drawers and never taken out again. An old Nokia. A school ID. A watch that stopped.

The mirror is on the wardrobe door, which is where an almirah mirror always is, and it is also the only chance you get to see V's face, since every other camera in the room is behind him. The reflection is real, rendered at half resolution and angled so it only ever catches the character. At this fidelity nobody will notice the shortcut.

The mirror also pairs with something on the fridge, which we will get to.

The fridge

The fridge is on the left wall below the wardrobe.

Inside there are three packets of milk, some fruit, and a leftover pizza in its box. Take a packet and there are two. Wait a moment, or close the door and open it again, and there are three. This is a deliberately naive trick with no inventory system behind it, and it is the right call: the only thing that matters is that the fridge feels stocked and the chai never stops working. Fruit and pizza restock the same way.

Opening the fridge in a dark room is also one of the nicest lighting moments in the game, that cold interior wedge cutting into all the orange.

On top sit a half-drunk water bottle whose level actually drops when you drink from it, a stack of bills and papers you can read up close, and a trimmer.

The bills are the most honest objects in the entire room. An electricity bill, an internet bill, a fee receipt. Nothing characterises a student's life faster than the paperwork of it.

The trimmer is the one that pays off elsewhere. Pick it up, carry it across the room to the wardrobe mirror, and it buzzes while V goes through the motions. Two objects that had nothing to do with each other now form a small daily routine, and unglamorous routine sells a lived-in room better than any amount of decoration.

On the door are alphabet magnets, scattered at random across the white. The letters are N, I, K, H, I, L. You can drag them anywhere you like, and most people will spend a minute rearranging them into something rude. Put them in the right order and something happens.

The kitchen corner

The bottom wall of the room is a small kitchen: a sink in the corner, a shelf running along the wall, an oven, the toaster, the induction cooktop, power outlets, and the usual scatter of steel utensils. Window one is above it.

The cooktop is where chai gets made. Pan on, water, milk from the fridge, tea, sugar, and then you wait while it comes up to the boil with the right sound and steam building over the pan. Then you pour.

The oven reheats the leftover pizza, which is the second cooking loop and a good excuse for another warm light source. The box empties as slices go.

The toaster pops with a genuinely satisfying mechanical thunk and produces toast that does nothing. It is there for the noise.

The sink runs water. It is also where used mugs go, and if you never wash them they pile up over a session, which is both funny and true.

The utensils are nudgeable and clatter. Flat-shaded steel at low polygon counts catches hard specular edges beautifully, so the cheapest props in the room are among the best-looking.

The power outlets charge the phone.

Chai

Chai is the signature interaction, and it is deliberately a sequence instead of a button.

Open the fridge, take a packet of milk, carry it to the cooktop, put the pan on, add everything, wait for it to boil, pour it into the mug. Pick up the mug and carry it wherever you want. Set it down and it leaves a ring. Drink it and it empties. Wash it in the sink or leave it there for later.

While you are holding a full cup, the room changes slightly. Steam rises off it and thins out over time. The colour grade warms a shade further. V stands differently. The room gets cosier while you hold it, which ties the interaction the whole game is built around directly to the light the whole game is built around.

The middle of the room

A Kashmiri durrie covers the centre of the floor, with a bean bag and the desk chair sitting on it.

The durrie does one thing that nobody will consciously notice: footsteps sound different on it. Soft on the weave, harder on the bare floor beside it. Surface-aware footsteps are invisible as a feature and enormous as a feeling.

The bean bag can be sat in and sinks under V's weight. **The chair** can be sat in, and it spins with real momentum that decays, and it can be shoved when it is empty. **The backpack** lies where it was dropped with notebooks spilling out of it, and it opens, and there are things inside.

The walls

The poster wall is the right-hand wall by the door, and it holds the map of Jammu and Kashmir, drawn as pixel art in the same PS1 register as everything else. It is a poster and only a poster. Nothing happens when you touch it, nothing zooms, nothing opens. It is on the wall for the same reason posters are on anyone's wall, and turning it into an interface would have made it an advertisement instead of a possession. Other posters go up beside it as they get chosen.

The whiteboard is drawable. You can draw on it with the cursor, rub it out, and draw again, and a reset button restores the real system design diagrams that were on it before you arrived. It costs almost nothing to build and people will lose several minutes to it.

The bookshelf holds programming, history, and geopolitics, and the books come out individually. Open one and there is a page about why it mattered. A shelf read book by book is a better account of a person than any list of skills, and it is the only place in the room where the history side of V sits alongside the code.

The boombox sits on the bookshelf, chunky enough to read clearly at low polygon counts. It plays music, you can skip tracks, and you can turn it up or down.

The fairy lights run along the shelf edge and switch on and off. When music is playing they pulse very gently with it.

The ceiling fan has an off position and three speeds, and this is where the room starts talking to itself: fan speed drives how much the curtains sway and whether loose papers on the desk lift at the corners. Objects affecting other objects is the whole difference between a room and a diorama.

The windows open. Open one and the outside gets louder and clearer and the breeze picks up and the curtains move more. Close it and everything muffles. **The curtains** draw shut, and shutting them at dusk kills the orange light and leaves you reaching for the lamp, which is a decorative object with a real consequence.

The room lights switch on and off, warm, layered on top of whatever the sun is doing.

The console sits wherever a console sits, with a cartridge you can push into it. It plays Snake. The cartridge label is the W4LLS mark.

The door

The door is on the right wall with a shoe rack beside it, chappals kicked off and one shoe knocked over.

The door is locked, and there is a note stuck to it. The room's name is scratched onto the frame, so the title exists as a physical object in the world and not only on a boot screen.

It is also the natural way to leave. Walking to a door and finding it shut is a better ending for a game with no ending than a close button in the corner of a browser.

A bathroom is part of the flat. Where it sits and what is in it has not been worked out yet, and that decision is deliberately being left until the main room is standing.

What the room does when you do nothing

Stand still and the room keeps moving.

The fan turns. The curtains breathe with it. Dust hangs visible in the shafts of sunset light coming through the window. Steam curls off the mug and fades. The phone screen lights up by itself with a notification and goes dark again. A chair that was spun three minutes ago is still very slowly coming to rest. At night, a moth works its way around the lamp.

V does things too. Leave him alone for twenty seconds and he stretches, checks his phone, yawns. Leave him for a minute and he walks over to the bed and sits down on his own.

That last one matters more than it sounds. A character who gets bored waiting for you is a character, and it is the cheapest personality you will ever buy.

Sound

The soundscape is tied to whichever of the four times of day is active. Birds at dawn. Traffic through the evening. Crickets at night. Rain against the glass when the weather rolled that way.

Underneath all of it, a wall clock ticks. At dusk, very faintly and from a long way off, the sounds of a Jammu evening, a temple bell, the call to prayer, held quiet enough to be atmosphere and never turned into a set piece.

Then the close sounds: keys clacking, the toaster, water in the sink, the fridge door, steel on steel, chai coming to the boil, a tennis ball on a wall.

And the headphones, which take all of it and push it away.

Things hidden

Nothing in the room is locked behind anything else, and none of these is required for anything.

There is a word you can type on the keyboard. There is an order the fridge magnets go in. There is a command the terminal answers. There is a file the dead hard drive gives up. There is a game inside the console. There are things in the wardrobe drawers, things inside books, things under the bed, things in the backpack, things on the USB sticks. Old phones, older photographs, a first line of code, a cricket ball with the seam gone.

The diary quietly notes what has turned up. Nothing rides on the number. It is a record, not a score, and the door stays shut either way.

Things the room remembers

Come back a week later and the room is roughly as you left it. The mug rings are still on the desk. The tennis ball is wherever it stopped. The plant has drooped again. The lights are how you left them, the magnets are how you arranged them, the boombox is on the track you last played, and the things you found are still found.

Persistence is cheap and it is the closest a browser tab gets to being a place you have been before.

Getting out fast

Not everyone wants to play a game, and some of the people least likely to want to are the ones this was built for.

So there is a small menu in the corner, styled to match but making no attempt to be part of the fiction, and it is there from the first frame. Resume, GitHub, LinkedIn, and a plain HTML

version of the portfolio with no 3D in it at all. The boot sequence can be skipped.

It slightly breaks the spell and it stays anyway. A recruiter with four minutes should never be trapped in someone's bedroom.

Behind the canvas, the page is a real static HTML document with actual text in it, so search engines and link previews get something to read. The preview image is the room at dusk.

Running well

Three quality levels, detected automatically, adjustable by hand. One real light in the scene, the sun; everything else that glows is faked with emissive materials, which is both faster and more period-accurate.

There is also a reduce motion setting, and what it does is turn down the vertex jitter. The PS1 wobble is constant sub-pixel movement across the entire frame, which is precisely the thing that makes motion-sensitive people feel ill. It is not an accessibility box being ticked. It is the specific accessibility problem this specific art style creates.

Built with

Three.js and React Three Fiber inside Next.js for the room itself. A custom shader stack for the look: vertex snapping, affine texture mapping, and a low-resolution render target scaled back up. Blender for the models. React layered over the canvas for the monitor desktop, the laptop, and anything else that becomes a screen. No server, no database, no backend of any kind. It is a static site and it deploys to Vercel or GitHub Pages.

On assets: the scope of this room is large and it is not being cut. The way it gets built in reasonable time is by buying good low-poly packs for everything generic, the furniture, the kitchen, the electronics, the clutter, and retexturing all of it onto one shared 128-pixel atlas with a single material so it comes out looking like it belongs to the same world. Everything personal gets built from scratch: V himself, the map poster, the diary, the sticky notes, the bills, the durrie, and the objects that are actually somebody's.

The retexture pass is what makes bought furniture stop looking bought.

Where this starts

The shader.

An empty room, four grey walls, and nothing in it except vertex jitter and affine texture warping, until it looks right. Everything else in this document is work that can be estimated. That first piece is the only part carrying real risk, and the entire identity of the project is resting on whether it lands. Better to find out in week one, standing in an empty grey box, than in week nine surrounded by fifty finished models.

After that: the room and V walking around it. Then the four lighting states. Then the monitor, because that is the part that has to work even if nothing else does. Then chai, then the fridge, then everything hanging off them. The toaster comes last and it is the first thing to go if time runs out.

Though it would be a shame. The toaster is very good.